Pavithraa Satchithanantham

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EDUCATION

Masters of Science, Computer Science — December 2025
Arizona State University, Tempe, Arizona — GPA: 3.93/4.00

Bachelor of Engineering, Electronics and Communication Engineering — May 2023
Ramaiah Institute of Technology, Bengaluru, India — GPA: 3.72/4.00

SKILLS

Python | Java | C | C++ | MLOps | Agentic AI | AWS | Artifical Intelligence | Machine Learning | LLM | PyTorch | TensorFlow | scikit-learn | HTML5 | CSS | JavaScript | SQL | MongoDB | React.js | Next.js | Node.js | REST | API | React Native | Web3 | Git | Linux | Agile | Automation | Unreal Engine | Object Oriented Programming | Data Structures and Algorithms

PROFESSIONAL EXPERIENCE

Software Engineering Intern — June 2025 – August 2025
EchoStar | Dish Network LLC — Denver, Colorado

- **Winner** of the prestigious **'Curiosity' CPAW Award** for demonstrating exceptional curiosity and initiative throughout internship.
- Pioneered 'JARVIS', an Agentic AI based Retrieval Augmented Generation System built on AWS achieving a stellar 95% quality index score enabling the DISH IT Video Services Team to query vast technical documentation and receive instant, accurate and updated information based on explicit and implicit knowledge on system architectures, service dependencies and workflows.
- Enhanced functionalities by designing a novel, role-aware and adaptive Developer and Executive 'Core' and 'Nexus' personas to support diverse user needs across technical and leadership teams.
- Increased team operational efficiency by up to 75% in targeted workflows by reducing SME dependency and enabling rapid issue resolution allowing engineers to shift focus from time consuming information retrieval to high value innovation.

Software Development Associate — June 2023 – November 2023
Hue Learn — Bengaluru, India

- Led a team of 7 developing a VR educational platform increasing user engagement by 70% over six months.
- Pioneered the multiplayer strategy for the platform using dedicated server technology and introduced 'crowd controlling' through the same to endow one or more players the ability to control and manipulate the actions of multiple characters within the game, and doubled the user base.
- Automated the generation of course content in the metaverse through HTTP and REST API request handling and JSON decoding, to fetch relevant content from the database and realize them in the metaverse as interactive 3D screens, 3D models and game worlds, with UI overlay from ReactJs.

Software Development Intern — April 2023 – May 2023
Hue Learn — Bengaluru, India

- Led a team of 3 conceptualizing and fabricating the look of characters to make more relatable character models helping the player connect with the character and also evoke strong emotions in them.
- Designed and constructed immersive game worlds featuring higher geometric realism in the metaverse using the Nanite technology, for the virtual reality educational platform leading to a 40% uptick in user experience.

Software Engineering Intern — October 2021 – February 2022
Curl — Bengaluru, India

- **Runner up** at the **(International) Sovereign Nature Winter Hackathon Challenge 2022**, Sovereign Nature Initiative (SNI) x De Ceuvel Hackathon, Feb 2022 and won a cash prize and tokens.
- Instrumental in developing Sattva a framework to enable an ecosystem to be self-sovereign built using Python Django along with an advanced Style GAN Model and an Object Detection Algorithm, to convert images created or owned by the park into unique and attractive artworks automatically.
- Minted artworks as NFTs and sold on marketplaces to generate revenue for self-sustenance of the park.

ADDITIONAL PROJECTS

GARUDA: An End-to-End Drone Detection and Management System — October 2021 – June 2022
Ramaiah Institute of Technology — Bengaluru, India

- **Winner** at **(National) Smart India Hackathon 2022**, Ministry of Education, Govt. of India, Aug 2022, won cash prize of $1200.
- Published a peer reviewed paper entitled ***"GARUDA: Third Eye for Detecting and Tracking Drones"*** at the ***2023 IEEE 2nd International Conference on Data, Decision and Systems (ICDDS)***.
- Published a peer reviewed paper entitled ***"A Deep Learning Approach to Classify Drones and Birds"*** at the ***2022 IEEE 2nd Mysuru Sub Section International Conference***.
- Employed a robust computer vision model capable of identifying small drones from long distances and low resolution CCTV footage with an impressive precision of 90.1% and a recall of 92.9%.
- Designed to track the status of the drones to infer if it is approaching, receding or moving laterally, and draw bounding boxes around the detected drones, update their status in a cloud database and trigger real-time alerts to personnel via mobile devices, to be installed in defense establishments such as DRDO (Defence Research and Development Organisation).